

Q33 - Kruskal vs Boruvka Algorithm (MST Comparison)

Aim: Construct MST using Kruskal and Boruvka algorithms and compare their performance.

Objectives:

- Construct MST using both algorithms.
- Compare edge selection and efficiency.

Kruskal Algorithm:

Select edges in increasing order while avoiding cycles using Union-Find.

Boruvka Algorithm:

Each component selects its cheapest outgoing edge in every iteration.

Comparison:

Kruskal works well for sparse graphs, while Boruvka is suitable for parallel processing and large graphs.

Complexity:

Kruskal: $O(E \log E)$

Boruvka: $O(E \log V)$

Scalability:

Boruvka scales better for distributed environments.

Conclusion:

Both algorithms efficiently generate MSTs with different strengths depending on graph characteristics.

Output Screenshot: Insert screenshot here.